

APPASA Strategy Document

Stage 4 – Analyze: Exploratory Data Analysis & Business Insights



APPASA: Analyze Stage

Introduction

I will use this **APPASA strategy document** to record my decisions and reflections as I work through **Stage 4: Analyze** of this project. This document will serve as a guide, helping me consider my responses and reflections throughout this stage of the data analysis process.

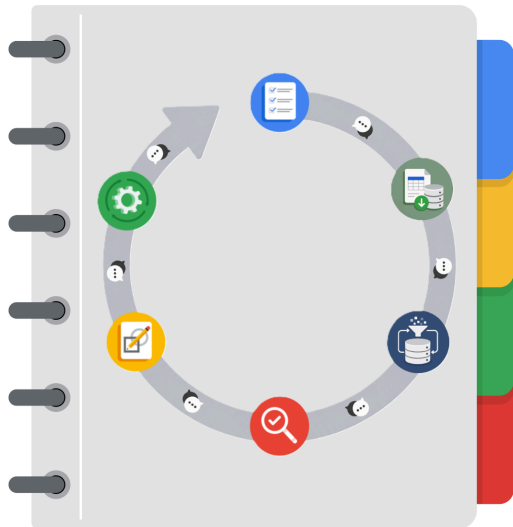
Portfolio Project Recap

Many of the goals I accomplished throughout the individual course portfolio projects are integrated into this end-to-end data analysis capstone project. These include:

- Creating a clear and structured project proposal.
- Demonstrating my understanding of SQL and relational database concepts.
- Using SQL to ingest, explore, clean, transform, validate, and organize data through structured queries and database operations.
- Organizing and analyzing a large-scale dataset to uncover meaningful insights and communicate the underlying business story.
- Developing a series of SQL scripts for ELT, feature engineering, data validation, and analytical querying, alongside a DuckDB-powered Jupyter Notebook documenting the complete APPASA methodology and reproducing the analytical workflow through SQL queries and visualizations.
- Computing descriptive statistics, performing exploratory and business-focused analyses, and creating analytical visualizations to support data interpretation and executive decision-making.
- Communicating findings through an executive summary, an interactive Power BI report, strategic business recommendations, and implementation-focused action plans for stakeholders.
- Translating analytical findings into evidence-based business recommendations, implementation strategies, success metrics, and future analytical opportunities through the final Act stage of the APPASA framework.

This capstone brings together the knowledge and technical skills I developed throughout the program, demonstrating my ability to execute a complete end-to-end data analysis project, from business problem definition to actionable business recommendations.

THE APPASA WORKFLOW



I will demonstrate to the company's marketing analytics team how I would apply the APPASA workflow to the upcoming Cyclistic Bike Share project. For each question presented in this APPASA strategy document, I will provide a structured and thoughtful response aligned with the corresponding stage of the APPASA framework: Ask, Prepare, Process, Analyze, Share and Act. This approach ensures clarity in my methodology, highlights my strategic planning skills, and illustrates a well-organized path from problem understanding to actionable insights.

Project Tasks

The following questions have been identified as essential to the Member vs Casual Rider Behavior Analysis Project. These questions are primarily situated within the Analyze stage of the APPASA workflow, focusing on exploring the processed dataset, identifying rider behavior patterns, discovering trends and relationships, and generating business insights. I will address each question by drawing on relevant materials from the SQL scripts, project notebook, the APPASA framework, and best practices in exploratory data analysis and business analytics.





Data Project Questions & Considerations

Analysis Environment

- What analytical environment and dataset were used for this stage?

The analytical phase was conducted using **MySQL** as the primary query engine and **MySQL Workbench** as the development environment for writing, executing, and documenting SQL queries. Unlike the Process stage, which focused on data validation, cleaning, transformation, and feature engineering, the Analyze stage operated exclusively on the finalized analytical dataset to perform descriptive statistics, comparative analyses, and business-focused exploration.

The dataset used throughout this stage was the **cyclistic_tripdata_2024_processed** table created during the Process stage. This processed analytical table contains **5,851,692** validated trip records and includes both the original trip attributes and the engineered features developed during data preparation, such as **ride_length_minutes**, **ride_date**, and **hour_of_day**. By using the processed table rather than the raw staging data, all analyses were performed on a clean, standardized, and quality-assured dataset.

The analytical workflow consisted of SQL aggregation functions, conditional logic, window functions, Common Table Expressions (CTEs), and analytical views to examine rider behavior across multiple dimensions, including rider type, ride duration, seasonality, day of the week, hour of the day, bike type, and station usage. This approach produced a reproducible, scalable, and well-documented analysis that directly addressed the project's primary business question while preparing the results for the subsequent Share stage through Power BI visualizations and an executive summary.

- Why was the processed analytical table used instead of the raw dataset?

The processed analytical table was used to ensure that all analyses were performed on a clean, validated, and standardized dataset rather than the raw imported data. During the Process stage, the raw trip records underwent comprehensive data validation, cleaning, transformation, feature engineering, and quality verification to resolve data quality issues and prepare the dataset for analysis. As a result, the processed table contained only reliable records suitable for generating meaningful business insights.

Using a separate processed analytical table also established a clear separation between data preparation and data analysis. The raw staging table remained preserved in its original state, ensuring data integrity and allowing every transformation to be reproduced or audited if necessary. Meanwhile, the Analyze stage could focus exclusively on exploring rider behavior and identifying trends without repeatedly applying cleaning or transformation logic within analytical queries.

This approach also improved the readability, maintainability, and reproducibility of the project. Engineered features such as **ride_length_minutes**, **ride_date**, and **hour_of_day** were already available in the processed dataset, enabling analytical queries to remain concise while ensuring that every result was generated from a consistent and quality-assured data source. This separation aligns with industry best



practices for analytical workflows, where data preparation and business analysis are treated as distinct stages within the data lifecycle.

Exploratory Analysis

- How was the analytical baseline established before investigating rider behavior?

Before conducting detailed rider behavior analyses, an analytical baseline was established to develop a comprehensive understanding of the processed dataset and verify that it was suitable for exploration. The analysis began by reviewing the overall size of the dataset, inspecting the table schema, and examining a representative sample of records. These initial checks confirmed the dataset structure, validated that the engineered features were available as expected, and provided familiarity with the analytical variables that would be used throughout the remainder of the project.

Following the dataset overview, descriptive statistics were calculated to summarize the ride duration variable across all processed trips. Key metrics, including the total number of rides, minimum ride duration, maximum ride duration, and average ride duration, provided an initial understanding of the dataset's overall characteristics before segmenting the data by rider type or temporal dimensions.

Establishing this analytical baseline ensured that subsequent analyses were interpreted within the broader context of the entire dataset rather than in isolation. It also provided a consistent reference point against which behavioral differences between annual members and casual riders, temporal usage patterns, bike preferences, and station-level trends could be compared throughout the Analyze stage.

- What descriptive statistics were calculated to understand the overall dataset?

To establish a foundational understanding of the processed analytical dataset, descriptive statistics were calculated for the **ride duration** variable across all processed trips. Specifically, the total number of rides, minimum ride duration, maximum ride duration, and average ride duration were computed using aggregate SQL functions. Together, these metrics provided a concise summary of the dataset's scale, range, and central tendency before conducting more detailed behavioral analyses.

The total ride count quantified the volume of observations available for analysis, while the minimum and maximum ride durations defined the overall range of valid trip lengths after the data cleaning process. The average ride duration served as an initial measure of central tendency, offering a high-level summary of typical trip length across the entire dataset.

Although the average provides a useful overall benchmark, it was not interpreted in isolation. Ride duration data is inherently susceptible to right-skewness due to a relatively small number of exceptionally long trips. Consequently, the average was treated as a descriptive reference point rather than a complete representation of typical rider behavior. This motivated the subsequent ride duration distribution analysis,



where rides were grouped into duration intervals to better understand the underlying distribution and assess how representative the mean truly was.

Rider Behavior Analysis

- How do annual members and casual riders differ in their overall riding behavior?

To establish the primary behavioral differences between rider groups, the processed dataset was first segmented by **member_casual** and analyzed using aggregate statistics. Three complementary analyses were performed: total ride volume, each rider type's proportional contribution to overall ride activity, and average ride duration.

The analysis revealed that annual members generated **3,706,732 rides (63.34%)**, while casual riders completed **2,144,960 rides (36.66%)** during the study period. This indicates that annual members account for nearly two-thirds of all trips, suggesting that they are the dominant contributors to Cyclistic's overall ride demand.

Behavioral differences also emerged when comparing average ride duration. Casual riders recorded an average trip length of approximately **20.92 minutes**, compared with **12.19 minutes** for annual members. This suggests that although annual members ride more frequently, casual riders typically spend considerably more time on each trip.

These findings established the first clear distinction between the two rider groups: **annual members exhibit higher ride frequency, whereas casual riders exhibit longer ride durations**. This initial comparison provided the analytical foundation for the subsequent analyses, which further examined how these behavioral differences vary across ride durations, temporal patterns, bike preferences, and station usage.

- How is ride duration distributed, and does the average accurately represent typical rider behavior?

To better understand rider behavior beyond the overall average, ride durations were grouped into meaningful time intervals and analyzed using both absolute ride counts and percentage distributions. Examining the distribution rather than relying solely on the mean provided a more complete understanding of how ride durations were concentrated across the dataset.

The analysis revealed a pronounced **right-skewed distribution**. Approximately **80.80%** of all rides were completed within **20 minutes**, with the largest concentration (**30.23%**) occurring in the **5–10 minute** interval. In contrast, only **2.50%** of rides lasted longer than **60 minutes**, indicating that exceptionally long trips were relatively uncommon.

The distribution was then segmented by rider type to compare behavioral patterns independent of differences in ride volume. Rather than comparing raw counts, percentages were calculated **within each rider type**, allowing the ride duration profiles of annual members and casual riders to be compared on an equal basis despite the larger number of member trips. This analysis showed that annual members were more

heavily concentrated in shorter-duration rides, whereas casual riders exhibited a greater proportion of medium- and long-duration trips. For example, **24.22%** of member rides lasted less than five minutes compared with **15.93%** of casual rides, while **5.66%** of casual rides exceeded one hour compared with only **0.66%** of member rides.

These findings demonstrate that the overall average ride duration should be interpreted with caution. Although the average provides a useful high-level summary, it does not accurately represent the typical rider experience because it is influenced by a relatively small number of long-duration trips. The ride duration distribution therefore provides a more representative view of rider behavior while reinforcing the conclusion that annual members primarily use Cyclistic for shorter, routine trips, whereas casual riders are more likely to undertake longer recreational rides.

Temporal Analysis

- How does rider activity vary throughout the year?

To understand how rider activity changes over time, the analysis examined trip patterns across both **monthly** and **seasonal** time periods. Monthly analysis provided a detailed view of fluctuations throughout the calendar year, while seasonal aggregation revealed broader trends associated with changing weather conditions and rider demand.

The results showed a clear seasonal pattern in Cyclistic usage. Ride activity increased steadily from winter into spring, reached its highest levels during the summer months, and gradually declined throughout autumn before reaching its lowest levels during winter. This progression reflects the strong influence of seasonal conditions on bike-share usage, with warmer weather encouraging substantially higher riding activity.

To support seasonal analysis, a reusable analytical view was created to classify each trip into one of four meteorological seasons based on its start month. This enabled ride volume, rider composition, and average ride duration to be analyzed consistently across seasons without repeatedly applying classification logic within individual queries.

Overall, the temporal analysis demonstrates that Cyclistic experiences predictable annual fluctuations in demand, with peak activity occurring during warmer months and reduced usage during colder periods. Establishing these yearly patterns provides important context for understanding when different rider groups are most active and supports the development of seasonally targeted marketing strategies in the subsequent stages of the project.

- How do annual members and casual riders differ across months and seasons?

To compare temporal riding patterns between rider groups, monthly and seasonal analyses were performed separately for annual members and casual riders. These analyses examined ride volume, rider composition, and average ride duration to identify how riding behavior changed throughout the year for each group.

The results showed that both rider types followed the same general seasonal cycle, with activity increasing during the warmer months and declining during colder periods. However, the magnitude of this variation differed considerably between the two groups. Annual members maintained relatively consistent riding activity throughout the year, suggesting that their trips were driven primarily by routine transportation and commuting needs. In contrast, casual riders exhibited much stronger seasonal fluctuations, with ride volumes increasing sharply during spring and summer before declining substantially during autumn and winter.

Differences were also evident in ride duration. While casual riders consistently recorded longer average trips than annual members across all months and seasons, both groups tended to take longer rides during the warmer months. This indicates that favorable weather not only encourages more trips but also supports longer riding sessions, particularly among casual users.

Overall, the analysis suggests that **annual members demonstrate stable, year-round riding behavior**, whereas **casual riders are considerably more influenced by seasonal conditions**. These findings reinforce the distinction between transportation-oriented usage by members and recreational usage by casual riders, providing valuable insight for designing marketing campaigns aimed at converting casual riders into annual members during periods of peak engagement.

- How does rider behavior change throughout the week?

To examine weekly riding patterns, trip activity was analyzed across all seven days of the week using ride volume, rider composition, and average ride duration. Together, these analyses provided insight into how rider behavior varies between weekdays and weekends.

The analysis revealed clear differences in weekly riding behavior between annual members and casual riders. Annual members generated the highest ride volumes on weekdays, particularly during the traditional workweek, indicating that their riding patterns are closely aligned with regular commuting and daily transportation. In contrast, casual riders were more active on weekends, when their share of total rides increased noticeably.

Ride duration analysis further reinforced these behavioral differences. Casual riders consistently recorded longer average trip durations than annual members throughout the week, with the longest rides typically occurring on weekends. Annual members, however, maintained relatively short and stable ride durations across all days, suggesting that their trips were primarily functional and time-efficient.

Overall, the weekly analysis highlights two distinct usage patterns: **annual members exhibit consistent weekday-oriented riding associated with routine travel**, whereas **casual riders demonstrate stronger weekend engagement and longer recreational trips**. These findings complement the monthly and seasonal analyses by showing that rider behavior varies not only throughout the year but also within a typical week, providing further evidence of fundamentally different usage motivations between the two rider groups.



- How does rider activity vary throughout the day?

To investigate daily riding patterns, trip activity was analyzed by **hour of the day** using ride volume, rider composition, and average ride duration. This analysis identified how demand fluctuates over a typical 24-hour period and revealed differences in riding behavior between annual members and casual riders.

The results showed that rider activity follows a distinct daily cycle, with relatively low ride volumes during the overnight hours, increasing activity throughout the morning, peak demand during daytime and early evening hours, and a gradual decline later in the night. These fluctuations indicate that bike-share usage is strongly influenced by typical daily routines and travel patterns.

Comparing rider types revealed notable differences in when trips occurred. Annual members exhibited pronounced peaks during traditional commuting hours, suggesting that many of their rides were associated with travel to and from work or other routine destinations. Casual riders, on the other hand, displayed a more evenly distributed pattern throughout the day, with higher activity during the afternoon and early evening when recreational and leisure travel is more common.

Average ride duration also varied throughout the day. Casual riders consistently recorded longer trips across most hours, while annual members maintained shorter and more stable ride durations. This pattern further supports the distinction between transportation-oriented usage among members and leisure-oriented usage among casual riders.

Overall, the hourly analysis demonstrates that **rider activity is shaped by daily travel routines**, with annual members exhibiting clear commuter-driven demand and casual riders showing greater flexibility in both ride timing and trip duration. These insights provide a more granular understanding of daily usage patterns and complement the broader temporal trends observed across weeks and seasons.

- What temporal patterns emerge when day of the week and hour of the day are analyzed together?

To obtain a more detailed understanding of rider activity, the day of the week and hour of the day were analyzed together using a two-dimensional temporal analysis. Examining these dimensions simultaneously made it possible to identify recurring demand patterns that would not be apparent when analyzing each variable independently.

The combined analysis revealed distinct temporal clusters in rider behavior. Annual members exhibited concentrated ride activity during weekday morning and evening commuting periods, reflecting consistent work-related travel patterns. In contrast, casual riders showed greater activity during weekend afternoons and early evenings, when recreational riding was most prevalent. These contrasting patterns indicate that the timing of trips is strongly influenced by the underlying purpose of travel.

The analysis also highlighted periods of relatively low demand, particularly during overnight hours across all days of the week. This consistent decline in activity reinforces the cyclical nature of bike-share usage and demonstrates how demand fluctuates according to both daily routines and weekly schedules.

Overall, the combined day-and-hour analysis provides a more comprehensive view of rider behavior than either dimension alone. By revealing **when** different rider groups are most active, it confirms that **annual**



members predominantly follow weekday commuting schedules, whereas **casual riders concentrate their activity during weekends and leisure hours**. These insights offer valuable guidance for operational planning, resource allocation, and the timing of marketing initiatives aimed at engaging casual riders during periods of peak recreational demand.

Usage Pattern Analysis

- How do bike type preferences differ between annual members and casual riders?

To understand differences in vehicle preferences, ride activity was analyzed across the available bike types for both annual members and casual riders. The analysis examined total ride volume, rider composition, and average ride duration for each bike type, providing insight into how different bicycles are used by each rider group.

The results showed that **classic bikes** were the most frequently used bike type overall and were preferred by both annual members and casual riders. However, clear differences emerged in the adoption of the remaining bike types. Annual members accounted for a larger proportion of rides across all bike categories, reflecting their higher overall usage of the bike-share system. Casual riders, while fewer in number, demonstrated relatively greater use of electric bikes, suggesting a preference for the additional convenience and reduced physical effort they provide during longer recreational trips.

Ride duration analysis further reinforced these behavioral differences. Regardless of bike type, casual riders consistently recorded longer average trip durations than annual members, indicating that the distinction between the two rider groups is driven more by **trip purpose** than by the bicycle itself. Annual members generally used each bike type for shorter, routine journeys, whereas casual riders tended to undertake longer leisure-oriented rides.

Overall, the bike type analysis indicates that **both rider groups favor classic bikes**, but **their usage patterns differ substantially**. Annual members primarily use bicycles for frequent, practical transportation, while casual riders are more inclined toward longer recreational trips and exhibit a comparatively stronger preference for electric bikes. These findings provide additional context for understanding rider behavior and support future marketing strategies aimed at encouraging casual riders to transition to annual memberships.

- Which stations and routes experience the highest rider activity?

To identify the most frequently used locations within the Cyclistic network, ride activity was analyzed at both the **station** and **route** levels. The analysis examined the most popular start stations, end stations, and origin–destination routes to determine where rider demand was most heavily concentrated and to understand overall travel patterns across the bike-share system.

The results showed that rider activity was concentrated around a relatively small number of high-demand stations. Several locations consistently ranked among the busiest as both trip origins and destinations,

indicating that they function as major transportation hubs within the network. The most frequently used routes connected these high-traffic stations, suggesting that riders regularly travel between key commercial, residential, and recreational areas.

Analyzing station usage also revealed that demand was not evenly distributed throughout the network. Instead, a subset of stations generated a disproportionately large share of all trips, while many others experienced comparatively lower activity. This concentration highlights the importance of strategically managing bicycle availability and station capacity at high-demand locations to maintain service reliability.

Overall, the station and route analysis demonstrates that **Cyclistic usage is centered around a core network of highly active stations and travel corridors**. Identifying these demand hotspots provides valuable operational insight for bicycle redistribution, station planning, and infrastructure investment, while also supporting marketing initiatives by revealing where rider engagement is naturally highest.

- How do station preferences differ between annual members and casual riders?

To compare spatial riding behavior, station usage was analyzed separately for annual members and casual riders. The analysis examined the most frequently used start and end stations for each rider type, allowing differences in location preferences to be identified independently of overall ride volume.

The results showed that annual members generated higher ride volumes across most stations, reflecting their greater overall use of the Cyclistic system. Their most frequently used stations were concentrated around major business districts, transit hubs, and densely populated urban areas, indicating that many member trips were associated with routine commuting and everyday transportation.

Casual riders, while using many of the same high-demand stations, displayed stronger preferences for stations located near parks, waterfronts, tourist attractions, and recreational destinations. This suggests that casual riders are more likely to begin and end trips in locations associated with leisure activities rather than daily commuting.

Despite these differences, several stations ranked highly for both rider groups, demonstrating that certain locations serve as central hubs regardless of trip purpose. However, the relative popularity of individual stations varied between annual members and casual riders, reflecting their distinct travel motivations and riding patterns.

Overall, the station preference analysis reinforces the broader findings of the project: **annual members primarily use the bike-share system for practical, transportation-oriented travel, whereas casual riders exhibit stronger preferences for recreational destinations and leisure-focused routes**. Understanding these spatial differences provides valuable insight for targeted marketing, bicycle redistribution strategies, and future infrastructure planning.



Business Insights

- What key trends and relationships were discovered throughout the analysis?

The analysis revealed several consistent trends that collectively explain how annual members and casual riders use the Cyclistic bike-share system. Across every analytical dimension, including rider composition, ride duration, temporal behavior, bike preferences, and station usage, the two rider groups demonstrated distinctly different usage patterns.

Annual members accounted for the majority of all rides and exhibited consistent riding behavior throughout the year, week, and day. Their trips were characterized by higher ride frequency, shorter average durations, pronounced weekday and commuting-hour activity, and frequent use of stations located near business districts and transit hubs. These patterns indicate that members primarily use Cyclistic as a reliable mode of everyday transportation.

In contrast, casual riders completed fewer trips but consistently recorded longer ride durations. Their activity was more strongly influenced by seasonality, weekends, and leisure hours, with greater engagement during warmer months and at stations near recreational and tourist destinations. Casual riders also demonstrated a comparatively stronger preference for electric bikes, suggesting a greater emphasis on convenience during longer recreational rides.

The ride duration analysis further revealed that trip lengths followed a right-skewed distribution, indicating that a relatively small number of long rides increased the overall average. As a result, distribution-based analyses provided a more representative understanding of rider behavior than average ride duration alone.

Taken together, these findings establish a clear relationship between rider type and trip purpose. Annual members exhibit stable, transportation-oriented behavior driven by routine travel, whereas casual riders display more variable, recreation-oriented behavior influenced by time, season, and location. These consistent behavioral distinctions formed the foundation for the business recommendations developed in the subsequent Share and Act stages.

- How do these analytical findings answer the project's primary business question?

The primary business question sought to understand **how annual members and casual riders differ in their riding behavior** in order to support strategies for converting casual riders into annual members. The analytical findings provide a clear and evidence-based answer by demonstrating that the two rider groups consistently exhibit distinct usage patterns across multiple dimensions of the bike-share system.

The analysis showed that annual members generate the majority of rides and display stable, transportation-oriented behavior characterized by frequent, shorter trips concentrated on weekdays, during commuting hours, and around business districts and transit hubs. In contrast, casual riders take fewer but longer trips, exhibit stronger seasonal and weekend activity, favor recreational locations, and show greater use of electric bikes. These differences remained consistent across temporal, behavioral, and spatial analyses,



indicating that the observed patterns reflect fundamentally different travel purposes rather than isolated trends.

By identifying these behavioral distinctions, the analysis provides a data-driven understanding of **who** the two rider groups are, **when** they ride, **where** they ride, and **how** they use the Cyclistic system. This comprehensive understanding directly addresses the project's primary business question and establishes the analytical foundation for the Share stage, where these findings are communicated through visualizations, and the Act stage, where targeted recommendations are developed to encourage casual riders to transition to annual memberships.